

Spinning top research — Solution

X25 (2025) — English translation

A1

0.20 pts

Write down in vector form the expression for the instantaneous speed $\vec{v}$ of fragment dm at the moment under consideration. Express your answer using $\vec{\Omega}$, $\vec{r}$.

We use Euler's formula for a rigid body.

Answer:

$$\vec{v} = [\vec{\Omega} \times \vec{r}]$$

A2

0.20 pts

Write down in vector form the expression for the angular momentum $d\vec{L}$ of the fragment dm at the moment under consideration. Express your answer using $\vec{\Omega}$, $\vec{r}$, dm .

By definition, the angular momentum of an element of mass dm :

Answer:

$$d\vec{L} = dm[\vec{r} \times [\vec{\Omega} \times \vec{r}]]$$

A3

0.30 pts

Obtain expressions for the components of the vector $d\vec{L} = (dL_x, dL_y, dL_z)$ in the coordinate system $OXYZ$. Express your answer using X , Y , Z , Ω_x , Ω_y , Ω_z , dm .

Let's expand the double vector product in the expression for $d\vec{L}$:

$$d\vec{L} = dm(\vec{r} \times (\vec{\Omega} \times \vec{r})) = dm(\vec{\Omega}r^2 - \vec{r}(\vec{\Omega} \cdot \vec{r})).$$

Let us write the obtained vector expression component-by-component:

$$dL_x = dm(\Omega_x(X^2 + Y^2 + Z^2) - X(\Omega_x X + \Omega_y Y + \Omega_z Z)), dL_y = dm(\Omega_y(X^2 + Y^2 + Z^2) - Y(\Omega_x X + \Omega_y Y + \Omega_z Z)), dL_z = dm(\Omega_z(X^2 + Y^2 + Z^2) - Z(\Omega_x X + \Omega_y Y + \Omega_z Z)).$$

Simplifying, we get the answer.

Answer:

$$dL_x = dm(\Omega_x(Y^2 + Z^2) - \Omega_y XY - \Omega_z XZ) dL_y = dm(-\Omega_x YX + \Omega_y(X^2 + Z^2) - \Omega_z YZ) dL_z = dm(-\Omega_x ZX - \Omega_y ZY + \Omega_z(X^2 + Y^2))$$

A4

0.50 pts

For arbitrary motion of a rigid body, we can write angular momentum in the form

$$L_x = I_{xx}\Omega_x + I_{xy}\Omega_y + I_{xz}\Omega_z;$$

$$L_y = I_{yx}\Omega_x + I_{yy}\Omega_y + I_{yz}\Omega_z;$$

$$L_z = I_{zx}\Omega_x + I_{zy}\Omega_y + I_{zz}\Omega_z.$$

Obtain expressions for the coefficients I_{xx} , I_{xy} , I_{xz} , I_{yx} , I_{yy} , I_{yz} , I_{zx} , I_{zy} , I_{zz} in the form of integrals over dm . The integral over dm is the sum over infinitesimal regions of a solid body of mass dm . For example, the coordinates of the center of mass can be written as

$$X_c = \frac{1}{m} \int X dm, \quad Y_c = \frac{1}{m} \int Y dm, \quad Z_c = \frac{1}{m} \int Z dm.$$

In further paragraphs, the formulas obtained here are not used.

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In further paragraphs the formulas obtained here are not used.

The answer is obtained by integrating over dm the expressions of the previous paragraph.

Answer:

$$I_{xx} = \int dm(Y^2 + Z^2) \quad I_{yy} = \int dm(X^2 + Z^2) \quad I_{zz} = \int dm(X^2 + Y^2) \quad I_{xy} = I_{yx} = - \int dmXY \quad I_{yz} = I_{zy} = - \int dmYZ \quad I_{zx} = I_{xz} = - \int dmZX$$

A5

0.50 pts

Derive expressions for $\vec{L}$ and T in the principal axes. Express your answer using $I_x, I_y, I_z, \omega_x, \omega_y, \omega_z, \vec{e}_x, \vec{e}_y, \vec{e}_z$.

In the principal axes, the expressions for the angular momentum components $\vec{L}$ take the form

$$L_x = I_x \omega_x, \quad L_y = I_y \omega_y, \quad L_z = I_z \omega_z.$$

This gives the answer.

According to the condition

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L}.$$

Using the expression for $\vec{L}$ found above, we arrive at the answer.

Answer:

$$\vec{L} = I_x \omega_x \vec{e}_x + I_y \omega_y \vec{e}_y + I_z \omega_z \vec{e}_z$$

A6

0.50 pts

Obtain expressions for the components of the vector $\dot{\vec{L}}$ in the moving coordinate system $(\dot{\vec{L}})_x, (\dot{\vec{L}})_y, (\dot{\vec{L}})_z$. Express your answer using $I_x, I_y, I_z, \omega_x, \omega_y, \omega_z, \dot{\omega}_x, \dot{\omega}_y, \dot{\omega}_z$.

Let us differentiate the previously obtained expression for $\vec{L}$. Since the unit vectors of the coordinate system are movable, they also need to be differentiated.

$$\dot{\vec{L}} = I_x \frac{d}{dt}(\omega_x \vec{e}_x) + I_y \frac{d}{dt}(\omega_y \vec{e}_y) + I_z \frac{d}{dt}(\omega_z \vec{e}_z) = I_x \dot{\omega}_x \vec{e}_x + I_y \dot{\omega}_y \vec{e}_y + I_z \dot{\omega}_z \vec{e}_z + I_x \omega_x \dot{\vec{e}}_x + I_y \omega_y \dot{\vec{e}}_y + I_z \omega_z \dot{\vec{e}}_z = I_x \dot{\omega}_x \vec{e}_x + I_y \dot{\omega}_y \vec{e}_y + I_z \dot{\omega}_z \vec{e}_z + \vec{\omega} \times \vec{L}$$

Let us write out the vector product in coordinates:

$$[\vec{\omega} \times \vec{L}] = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \omega_x & \omega_y & \omega_z \\ L_x & L_y & L_z \end{vmatrix} = \vec{e}_x(\omega_y L_z - \omega_z L_y) + \vec{e}_y(\omega_z L_x - \omega_x L_z) + \vec{e}_z(\omega_x L_y - \omega_y L_x)$$

Thus obtain projection $\dot{\vec{L}}$ on axis system $Oxyz$.

Answer:

$$(\dot{\vec{L}})_x = I_x \dot{\omega}_x + (I_z - I_y) \omega_y \omega_z \quad (\dot{\vec{L}})_y = I_y \dot{\omega}_y + (I_x - I_z) \omega_z \omega_x \quad (\dot{\vec{L}})_z = I_z \dot{\omega}_z + (I_y - I_x) \omega_x \omega_y$$

B1

0.40 pts

Find the projections of angular velocity $\omega_x, \omega_y, \omega_z$. Express your answer using $\omega_s, \dot{\theta}, \dot{\varphi}, \theta$.

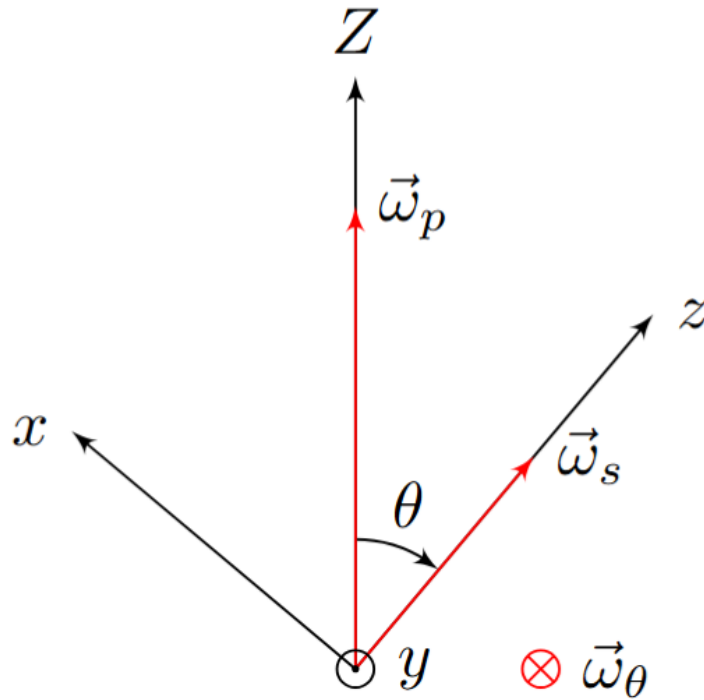
Projecting $\vec{\omega}_s, \vec{\omega}_\theta, \vec{\omega}_\varphi$ onto the axes x, y, z and taking into account

$$\vec{\omega}_\theta = -\dot{\theta}\vec{e}_y, \quad \vec{\omega}_\varphi = \dot{\varphi}\vec{e}_Z,$$

we get the answer.

Answer:

$$\omega_x = \dot{\varphi} \sin \theta \omega_y = -\dot{\theta} \omega_z = \omega_s + \dot{\varphi} \cos \theta$$



B2

0.40 pts

Show that the projection L_1 of the angular momentum $\vec{L}$ onto the fixed vertical axis OZ is constant. Show that the projection L_2 of the angular momentum $\vec{L}$ onto the moving axis Oz is constant.

The moment of gravity is expressed by the formula

$$\vec{M} = \vec{l} \times m\vec{g},$$

from which follow, that $\vec{M}$ perpendicular plane OZz . From the equation of moment in the stationary axes it follows

$$\dot{L}_1 = \dot{L}_Z = M_Z = 0.$$

From Euler's equation for the Oz axis it follows

$$I_z \dot{\omega}_z + (I_y - I_x) \omega_x \omega_y = 0.$$

due to the symmetry $I_x = I_y$, therefore

$$\dot{L}_2 = I_z \dot{\omega}_z = 0.$$

Note that for perpendicularity of $\vec{M}$ and Oz for conservation L_2 is not enough.

B3

0.60 pts

Using the results of step A5, express the values of the projections L_1 and L_2 in terms of ω_s , $\dot{\theta}$, $\dot{\varphi}$, θ , A , C .

Let us express L_1 through the projections $\vec{L}$:

$$L_1 = L_z \cos \theta + L_x \sin \theta = A\omega_x \sin \theta + C\omega_z \cos \theta.$$

Using the result of part B1, we obtain the final answer for L_1 :

$$L_1 = C(\omega_s + \dot{\varphi} \cos \theta) \cos \theta + A\dot{\varphi} \sin^2 \theta.$$

For L_2 correct

$$L_2 = L_z = C\omega_z = C(\omega_s + \dot{\varphi} \cos \theta).$$

Answer:

$$L_1 = C(\omega_s + \dot{\varphi} \cos \theta) \cos \theta + A\dot{\varphi} \sin^2 \theta, L_2 = C(\omega_s + \dot{\varphi} \cos \theta)$$

B4

0.70 pts

At this point we will assume that $l = 0$, and the moment of gravity can be ignored. Let the angular momentum of the top is equal to L_0 and directed along the axis OZ . At time $t = 0$ the values of angles $\theta = \theta_0$, $\varphi = \varphi_0$. Find the dependence of angles θ , φ on time. This movement is called regular precession.

In the described case

$$L_1 = L_0, \quad L_2 = L_0 \cos \theta.$$

due to the constancy of the quantities L_0 , L_1 , L_2 , the angle θ also remains constant. Using the expressions for L_1 , L_2 from previous part, we get the system of equations:

BEEQUATION

Solve system, obtain

$$\dot{\varphi} = \frac{L_0}{A}$$

Further dependence φ from time to time turns out to be simple integration.

Answer:

$$\theta(t) = \theta_0, \varphi(t) = \varphi_0 + \frac{L_0}{A}t$$

C1

0.60 pts

For an arbitrary position of the top, express the total energy E of the motion of the top (the sum of kinetic and potential) in terms of ω_s , $\dot{\theta}$, $\dot{\varphi}$, A , C , m , g , l , θ .

Let's write down the expressions for potential and kinetic energies:

$$T = \frac{1}{2}(C\omega_z^2 + A(\omega_x^2 + \omega_y^2)) = \frac{1}{2}(C(\omega_s + \dot{\varphi} \cos \theta)^2 + A(\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta)), \Pi = mgl \cos \theta.$$

Total energy:

$$E = T + \Pi.$$

Answer:

$$E = \frac{1}{2}(C(\omega_s + \dot{\varphi} \cos \theta)^2 + A(\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta)) + mgl \cos \theta$$

C2

0.30 pts

Express ω_s and $\dot{\varphi}$ in terms of L_1 , L_2 , A , C , θ using the results from step B3.

The answer is obtained by solving the system for ω_s , $\dot{\varphi}$.

$$\begin{cases} L_1 \ddot{\varphi} = \dot{\varphi}(A \sin^2 \theta + C \cos^2 \theta) + C \omega_s \cos \theta \\ L_2 \ddot{\theta} = C(\omega_s + \dot{\varphi} \cos \theta) \end{cases} \quad (1)$$

Answer:

$$\dot{\varphi} = \frac{L_1 - L_2 \cos \theta}{A \sin^2 \theta} \omega_s = \frac{L_2}{C} - \frac{L_1 - L_2 \cos \theta}{A \sin^2 \theta} \cos \theta$$

C3

0.60 pts

Using the constancy L_1 , L_2 , reduce the expression for the total energy E to the form

$$E = \frac{\mu}{2} \dot{\theta}^2 + U(\theta).$$

Obtain expression for μ , $U(\theta)$ through A , C , L_1 , L_2 , m , g , l . Note. As is known, the total energy is determined up to a constant, which for convenience can be set equal to zero.

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Substitution into the expression for the energy ω_s and $\dot{\varphi}$ from the previous paragraph leads to the result

$$E = \frac{A}{2} \dot{\theta}^2 + \frac{(L_1 - L_2 \cos \theta)^2}{2A \sin^2 \theta} + mgl \cos \theta + \frac{L_2^2}{2C}.$$

The last constant term can be discarded:

$$E = \frac{A}{2} \dot{\theta}^2 + \frac{(L_1 - L_2 \cos \theta)^2}{2A \sin^2 \theta} + mgl \cos \theta.$$

The resulting expression yields the answer.

Answer:

$$\mu = AU(\theta) = mgl \cos \theta + \frac{(L_1 - L_2 \cos \theta)^2}{2A \sin^2 \theta}$$

D1

0.80 pts

Find the period of oscillation of the top. You will receive the answer in the form of a definite integral containing A , C , l , θ_0 , g . Also obtain a simplified expression for the case $\pi - \theta_0 \ll 1$.

For given initial conditions

$$L_1 = L_2 = 0.$$

The law of energy conservation in this case takes the form

$$\frac{A \dot{\theta}^2}{2} + mgl \cos \theta = mgl \cos \theta_0.$$

Hence be expressed dependence $\dot{\theta}(\theta)$:

$$\dot{\theta} = \pm \sqrt{\frac{2mgl}{A} (\cos \theta_0 - \cos \theta)}.$$

Motion from $\theta = \theta_0$ to $\theta = \pi$ with $\dot{\theta} > 0$ takes a a quarter period. To find this time, we need to integrate the equation

$$dt = \frac{d\theta}{\dot{\theta}}.$$

Integrate:

$$\frac{T}{4} = \sqrt{\frac{A}{2mgl}} \int_{\theta_0}^{\pi} \frac{d\theta}{\sqrt{\cos \theta_0 - \cos \theta}}.$$

Let's introduce the variable $\beta = \pi - \theta$, then $\beta \ll 1$. Method 1 Taking into account the smallness of β , the expression for energy will be written as:

Such form equation corresponds to harmonic oscillations with period

2 method Introduce $\beta_0 = \pi - \theta_0$. Transform integral

in consider approximation.

Substituting the found value into the original expression for T leads to the answer obtained method 1.

Taking into account the smallness of β , the expression for energy will be written in the form:

Such form equation corresponds to harmonic oscillations with period

Let's enter $\beta_0 = \pi - \theta_0$. Let's transform the integral

in the considered approximation.

Substituting the found value into the original expression for T results in the answer obtained in method 1.

$$T = 4\sqrt{\frac{A}{2mgl}} \int_{\theta_0}^{\pi} \frac{d\theta}{\sqrt{\cos \theta_0 - \cos \theta}}$$

D2

0.20 pts

Express L_1 and L_2 in terms of A , C , ω_0 .

In a given situation $L_1 = L_2$. Their values are instantly obtained from the expression $\vec{L}$ in the principal axes.

Answer:

$$L_1 = L_2 = C\omega_0$$

D3

0.50 pts

Find a condition relating A , C , m , g , l , ω_0 such that the position $\theta = 0$ is stable for the deviations described above. This condition is called Maievsky's condition. It turns out that if the found condition is satisfied, the equilibrium position is stable even if the conservation condition L_1 and L_2 is abandoned.

At $L_1 = L_2 = C\omega_0$ the expression for the energy becomes

$$E = \frac{A}{2}\dot{\theta}^2 + mgl \cos \theta + \frac{C^2\omega_0^2(1 - \cos \theta)^2}{2A \sin^2 \theta}.$$

Expand taking into account smallness θ :

$$E = \frac{A\dot{\theta}^2}{2} + \left(\frac{C^2\omega_0^2}{4A} - mgl \right) \frac{\theta^2}{2}.$$

For stability it is necessary that the coefficient before θ^2 was positive, which determines the required condition.

Answer:

$$\frac{C^2\omega_0^2}{4A} > mgl$$

D4

0.70 pts

Obtain an equation for θ that is satisfied by the angle θ_1 . The equation may contain A , C , L_1 , L_2 , m , g , l . Clue. $U(\theta)$ can be represented as a function of the new variable $s = \cos \theta$, which greatly simplifies the solution.

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In the equilibrium position $\frac{d}{d\theta}U(\theta) = 0$. Let's move on to the variable $s = \cos \theta$. Then

$$\frac{d}{d\theta}U(\theta) = \frac{d}{ds}U(s) \cdot \frac{ds}{d\theta} = \frac{d}{ds}U(s) \cdot \sin \theta.$$

Hence it follows, that condition

$$\frac{d}{d\theta}U(\theta) = 0$$

is equivalent to the condition

$$\frac{d}{ds}U(s) = 0$$

for all $\theta \in (0, \pi/2)$. Obtain $\frac{d}{ds}U(s)$

$$U(s) = mgl s + \frac{(L_1 - L_2 s)^2}{2A(1 - s^2)} \frac{d}{d\theta}U(s) = mgl - \frac{(L_1 - L_2 s)L_2}{A(1 - s^2)} + \frac{(L_1 - L_2 s)^2}{A(1 - s^2)^2}.$$

Simplify expression and transition inversely to variable θ . Equating it to zero, we obtain the required equation.

Answer:

$$\frac{(L_2 \cos \theta - L_1)(L_2 - L_1 \cos \theta)}{\sin^4 \theta} + Amgl = 0$$

D5

0.80 pts

Find the possible precession rates of the top ω_p at $\theta = \theta_1$. Express your answer using $A, C, m, g, l, \omega_s, \theta_1$. One of the solutions can be to substitute expressions for L_1, L_2 into the equation from D4 through the angular velocities ω_s, ω_p .

Substituting the expressions for L_1 and L_2 from point B3 into the equation obtained in the previous point, we arrive at a quadratic equation for ω_p :

$$\omega_p^2(A - C) \cos \theta_1 - \omega_p C \omega_s + mgl = 0.$$

It has two solutions, which is the answer.

Answer:

$$\omega_p = \frac{C\omega_s}{2(A - C) \cos \theta_1} \left(1 \pm \sqrt{1 - \frac{4mgl(A - C) \cos \theta_1}{C^2\omega_s^2}} \right)$$

D6

0.30 pts

Obtain approximate expressions for the frequencies in the case of $mgl \ll C\omega_s^2$. Consider that A and C are quantities of the same order.

Let us expand ω_p using the approximate formula $\sqrt{1+x} \simeq 1 + \frac{x}{2}$ for $x \ll 1$. For a larger root:

For a smaller root:

Note that the frequency ω_p^{slow} coincides with the precession frequency of the “fast” top in the simplified model, that is, when the contribution of precession to the angular momentum is neglected.

Answer:

$$\omega_p^{fast} = \frac{C\omega_s}{(A - C) \cos \theta_1}$$

D7

0.70 pts

Determine the angle α_{max} of the maximum deviation of the top axis from the horizontal during the subsequent movement. Express your answer in terms of A, C, m, g, l, ω_0 .

Let us write the expression for energy taking into account the initial conditions:

$$L_1 = 0, L_2 = C\omega_0. E = \frac{A\dot{\theta}^2}{2} + mgl \cos \theta + \frac{C^2\omega_0^2 \cos^2 \theta}{2A \sin^2 \theta} = 0.$$

Transition to variable α by formula:

$$\theta = \frac{\pi}{2} + \alpha.$$

Then

$$\frac{A\dot{\alpha}^2}{2} - mgl \sin \alpha + \frac{C^2\omega_0^2 \sin^2 \alpha}{2A \cos^2 \alpha} = 0.$$

Condition $\alpha = \alpha_{max}$ correspond $\dot{\alpha} = 0$. Substitute $\dot{\alpha} = 0$ in the equation above and transforming, we obtain a quadratic equation on sine maximum angle deviation:

$$\sin^2 \alpha_{max} + \frac{C^2\omega_0^2}{2Amgl} \sin \alpha_{max} - 1 = 0.$$

Solve obtain square equation:

$$\sin \alpha_{max} = \pm \sqrt{1 + \left(\frac{C^2\omega_0^2}{4Amgl} \right)^2} - \frac{C^2\omega_0^2}{4Amgl}$$

Since $|\sin \alpha_{max}| \leq 1$, follow choose root with «+».

Answer:

$$\alpha_{max} = \arcsin \left(\sqrt{1 + \left(\frac{C^2\omega_0^2}{4Amgl} \right)^2} - \frac{C^2\omega_0^2}{4Amgl} \right)$$

D8

0.70 pts

Find the dependence of the angle of deviation of the top axis from the horizontal on time $\alpha(t)$. Express your answer in terms of A, C, m, g, l, ω_0 .

Let us write an approximate expression for α_{max} in this case:

$$\alpha_{max} = \frac{2Amgl}{C^2\omega_0^2} \sim \frac{mgl}{C\omega_0^2} \ll 1.$$

in force smallness α_{max} can consider expression for energy for small α . Taking into account initial condition total energy equal zero.

$$\frac{A\dot{\alpha}^2}{2} + \frac{C^2\omega_0^2}{2A}\alpha^2 - mgl\alpha = 0$$

Such form expression for energy correspond oscillation with frequency

$$\omega = \frac{C}{A}\omega_0,$$

amplitude

$$\alpha_0 = \frac{Amgl}{C^2\omega_0^2}$$

and displacement from center

$$\alpha_1 = \alpha_0 = \frac{Amgl}{C^2\omega_0^2}.$$

Taking into account all find parameter motion and initial condition obtain dependence $\alpha(t)$.

Answer:

$$\alpha(t) = \frac{Amgl}{C^2\omega_0^2} \left(1 - \cos\left(\frac{C}{A}\omega_0 t\right) \right)$$

D9

0.50 pts

Find the dependence of the precession angle on time $\varphi(t)$. At the initial moment $\varphi(0) = \varphi_0$. Express your answer using $A, C, m, g, l, \omega_0, \varphi_0$.

Let's express $\dot{\varphi}(t)$ using the found dependence $\alpha(t)$:

$$\dot{\varphi} = \frac{L_1 - L_2 \cos \theta}{A \sin^2 \theta} = \frac{C\omega_0 \sin \alpha}{A \cos^2 \alpha} \simeq \frac{mgl}{C\omega_0} \left(1 - \cos\left(\frac{C}{A}\omega_0 t\right) \right)$$

Integrating, we get the answer.

Answer:

$$\varphi(t) = \varphi_0 + \frac{mgl}{C\omega_0} \left(t - \frac{A}{C\omega_0} \sin\left(\frac{C}{A}\omega_0 t\right) \right)$$

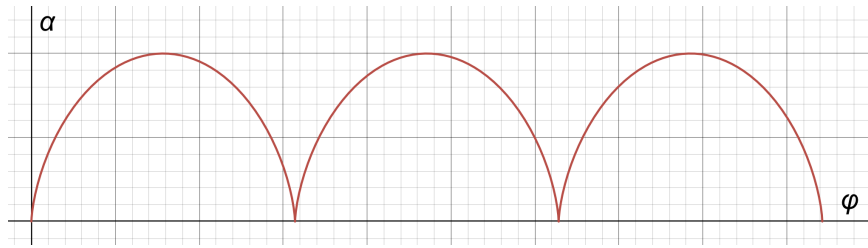
D10

0.50 pts

Draw a characteristic graph of the dependence $\alpha(\varphi)$. Determine the angle $\Delta\varphi$ through which the top axis rotates between two successive returns of the top axis to a horizontal position. Express your answer using A, C, m, g, l, ω_0 .

Expressions $\varphi(t), \alpha(t)$ parametrically define a cycloid, the characteristic form of which is presented in the figure below.

The time between subsequent instants $\alpha = 0$ is $t = \frac{2\pi A}{\omega_0 C}$. Substituting t into the expression for φ gives the answer.



D11

0.50 pts

Calculate $\Delta\varphi$ for the described system.

Let's find A , C , l :

$$C = \frac{(2m)R^2}{2} = mR^2 A = \frac{(2m)R^2}{4} + (2m)R^2 + \frac{mR^2}{3} = \frac{17}{6}mR^2 l = \frac{2mR + m\frac{R}{2}}{3m} = \frac{5}{6}R$$

Substitute in formula for $\Delta\varphi$ find quantity (is required also account for, that the mass of the top is $3m$) and get the answer.

Answer:

$$\Delta\varphi = \frac{17}{120}\pi$$